

Modeling Setup

- First, the target was divided into two classes: 0 = Low Severity and 1 = High Severity.
- Then, the dataset was split into three parts: 70% Training, 15% Validation, and 15% Testing.
- The Training data was used to teach the model.
- The Validation data was used to check if the model was learning correctly.
- Finally, the Testing data was used to test the model's performance.

Models Trained

- A total of 10 machine learning models were used in this study.
- These models were trained using the same dataset.
- Some of the models are Random Forest, XGBoost, LightGBM, CatBoost, and Decision Tree.
- AdaBoost and other machine learning models were also included.
- The performance of all models was compared to find the best one.

Model Filtering Rules

- Rule 1
 - Models with accuracy below 80% were removed.
- Rule 2
 - Models with accuracy above 97% were removed because they may be overfitting the data.
- Rule 3
 - Models with a Train–Test accuracy gap greater than 5% were removed because they may have memorized the training data instead of learning general patterns.
- Result

- Out of the 10 models, 9 passed the selection criteria.
- Only the MLP Classifier was excluded.
- Its training accuracy was 90.83%, while its testing accuracy was 85.60%.
- The accuracy gap was 5.23%, which is greater than the 5% limit.
- Therefore, the MLP Classifier was removed.

Full Model Results

- This table shows the results of all machine learning models.
- LightGBM achieved the best performance with a test accuracy of 94%.
- XGBoost ranked second with a test accuracy of 93.53%.
- CatBoost ranked third with a test accuracy of 93.53%.
- These models performed better than the other models in the study.

Final Model Comparison

- Accuracy
 - Accuracy shows how many predictions the model made correctly.
 - It is measured out of 100 predictions.
 - If the accuracy is 94%, the model predicts 94 out of 100 cases correctly.
- Precision
 - Precision shows how often the model is correct when it predicts "High Severity."
 - It measures the accuracy of the model's positive predictions.
 - Higher precision means fewer false alarms.
- Recall
 - F1-Score combines both Precision and Recall into a single value.
 - It measures the overall balance between Precision and Recall.
 - A higher F1-Score indicates better overall model performance.
- ROC-AUC

- ROC-AUC measures how well the model separates the two classes.
- The closer the ROC-AUC value is to 1.0, the better the model performs.
- LightGBM achieved a ROC-AUC of 97.2%, indicating excellent prediction performance.
- Confusion Matrix
 - The Confusion Matrix is the model's prediction report card.
 - It shows how many predictions are correct and how many are incorrect.
 - For example, if there are 100 accidents, the model may predict 94 correctly and 6 incorrectly.
- ROC Curve
 - This is the ROC curve.
 - If the curve goes closer to the top-left corner, the model performs better.
 - If it is close to the diagonal line, the model's predictions are not very good.
- Feature Importance
 - Feature Importance shows which features help the model the most.
 - It identifies the most important columns for making predictions.
 - The higher the importance, the more that feature helps the model.

SHAP Analysis

- SHAP explains why the model made a prediction.
- It shows how each feature affects the prediction.
- SHAP helps us understand the model instead of treating it as a black box.
- SHAP identified Speed_Violation_Risk_Score as the most important feature.

- This means that higher speed-related risk increases the chance of a more severe accident.
- This is the main finding of the thesis.

Learning Curve

- The Learning Curve shows whether the model is learning well or overfitting.
- In this study, the Training and Validation curves were close to each other.
- This indicates that the model is not overfitting and is expected to perform well on new data.

Final Conclusion

- Finally, the study identified LightGBM as the best-performing model.
- It achieved 94% Accuracy, 97.2% ROC-AUC, and showed no overfitting.
- SHAP identified Speed_Violation_Risk_Score as the most important feature.
- This means that higher speed-related risk increases the chance of a high-severity accident.
- Therefore, the main conclusion of this thesis is that accidents with higher speed violation risk are more likely to be severe.